



# Change One Script

Alter one move and describe what changes

Name: \_\_\_\_\_

Date: \_\_\_\_\_

★ I can change one sprite's script and describe how it changes whether they meet.

## Change it and see

Bit (green) and Pixel (purple) both start on the green flag ►. First run them as they are. Then we will CHANGE one block in Pixel's script and run again. Watch how changing one move changes who is ahead and whether they meet.

## The number path



## Bit's script

► start at 0

forward 3

## Pixel's script NOW

► start at 5

back 1

### 1 Before the change

Run both as they are. Bit: \_\_\_\_ Pixel: \_\_\_\_\_. Do they meet? Circle YES or NO. Who is ahead?

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## The change

Now change Pixel's move from 'back 1' to 'back 2'. Bit's script stays the same. Here is Pixel's NEW script:

## Pixel's script AFTER the change

▶ start at 5

back 2

### 2 After the change

Run both again with Pixel's NEW script. Bit: \_\_\_\_ Pixel: \_\_\_\_\_. Do they meet now? Circle YES or NO.

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### 3 Describe the change

Write one sentence: how did changing Pixel's one move change what happens when both run?

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## Big idea

Changing one block in one script can change the whole result. Here making Pixel go back 2 instead of back 1 moved Pixel from 4 to 3 — so now Pixel meets Bit instead of staying ahead.



# Change One Script

Quick facilitation notes & answer key

## What this worksheet teaches

Students change one move in ONE character's set and describe how it changes the COMBINED result — whether the two now meet. Changing one part and explaining the new outcome is the heart of Grade 2 coding. No coding experience needed.

## Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: put two counters on a number line 0 to 5 — green for Bit, purple for Pixel — and move BOTH at the same time so children see the two sets of moves run side by side.

## How to lead it (about 20 minutes)

1. Show them (you do). Run both as they are: they do not meet; Pixel is ahead.
2. Do one together. Exercise 1 — record both landings and who is ahead.
3. Let them try. Students run with Pixel's new move (Ex 2), then write how the one change changed the result (Ex 3).

## Answer key (these are the verified correct answers)

Exercise 1 (before) — Bit: 0, forward 3 → 3. Pixel: 5, back 1 → 4. They do not meet; Pixel (4) is ahead of Bit (3).

Exercise 2 (after) — Pixel's new move: 5, back 2 → 3. Bit still 3. Both on 3 → YES, they meet.

Exercise 3 — sample: "Changing Pixel to back 2 made Pixel land on 3 instead of 4, so now Pixel meets Bit instead of being ahead." Accept any answer linking the change to the new outcome.

## If students get stuck — and ways to adjust

Watch for: re-running Bit too. Only Pixel's move changed; Bit's result stays the same.

Make it easier: compare the two landing numbers before and after the change.

Make it harder: ask what change to BIT (instead of Pixel) would also make them meet.

## Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, change one, and explain what happens when both run together.

## You'll know it worked when

The child runs both before and after, and explains how Pixel's one change flipped the result to "meet".